

BRAC UNIVERSITY

Department of Computer Science and Engineering

Smart Bangladesh Digital City Network

Design and Implementation Report

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1 Assumptions and Design Decisions

The specification leaves several points open. Each decision below is stated together with the reason it was taken, so that the design can be assessed on its reasoning and not only on its output.

1.1 A1 — The PHD–TMA link is excluded from RIP

The specification requires a direct PHD–TMA link and also requires that a failure of the SCCC–TMA path reroute traffic through $\text{TMA} \rightarrow \text{DCS} \rightarrow \text{ERU} \rightarrow \text{SCCC}$. These two clauses conflict. If RIP runs across the PHD–TMA serial, then when TMA loses the central switch it simply relearns every network from PHD at administrative distance 120, the floating static at distance 130 never installs, and the required backup path is never taken.

The link is therefore cabled, addressed and reachable, but is declared `passive-interface` in RIP on *both* routers. Its subnet is still advertised to the rest of the network, while no routing information crosses it. The physical redundancy the brief asks for exists, and the failover path remains deterministic.

1.2 A2 — SCCC redistributes its static routes into RIP

SCCC is required to run both dynamic and static routing and to know every network. On its own this does not let TMA and PHD, which are RIP-only, learn the ERU, DCS and ESC networks. The command `redistribute static metric 5` under `router rip` on SCCC is what makes the requirement that TMA reach all external networks through SCCC true in practice.

1.3 A3 — DNS, and the absence of MX records in Packet Tracer

A single DNS server at SCCC is authoritative for all zones and is handed to every host by DHCP. Packet Tracer implements no MX record type; when a mail server relays a message addressed to another domain it performs a plain A lookup on the bare domain name. Records for `tma.gov.bd` and `esc.gov.bd` are therefore published in addition to the `mail.*` names, and both mail servers are configured with 11.8.1.250 as their own DNS server. Without this, mail inside each domain works and cross-domain mail fails.

1.4 A4 — Address reservation convention

In every LAN, host addresses .1 to .9 are reserved for the gateway and future infrastructure, and the top of the range is reserved for servers and printers. Both bands are excluded from the DHCP pools, so a dynamically assigned address can never collide with a static one.

1.5 A5 — Laboratory credentials

All routers use `enable secret cisco123` and line password `cisco`; all mail accounts use password `cisco123`.

2 VLSM Design

2.1 Base network

Student ID 22201108 $\rightarrow$ last four digits 1108 $\rightarrow$ pairs 11 and 08 $\rightarrow$ base network 11.8.0.0/16, mask 255.255.0.0, giving 65 536 addresses.

2.2 Sizing the subnets

Each requirement is rounded up to the next power of two after adding two addresses for the network and broadcast addresses. Blocks are then allocated largest-first, so that a small subnet can never block a large one. Table 1 shows the calculation.

Table 1: Host requirement to block size

Order	Network	Hosts	+2	Block	Prefix	Usable	Spare
1	SCCC LAN	320	322	512	/23	510	190
2	DCS LAN	260	262	512	/23	510	250
3	PHD LAN	220	222	256	/24	254	34
4	TMA LAN	180	182	256	/24	254	74
5	ESC LAN	140	142	256	/24	254	114
6	ERU LAN	100	102	128	/25	126	26
7	Central segment	3	5	8	/29	6	3
8–12	WAN links (5 of)	2	4	4	/30	2	0

2.3 VLSM allocation tree

Figure 1 shows how 11.8.0.0/16 is divided down to the six department LANs. The whole design fits inside 11.8.0.0/21; everything from 11.8.8.0 upward is left untouched for growth. Figure 2 expands the infrastructure block.

2.4 Subnet summary

Table 2: Complete VLSM allocation from 11.8.0.0/16

Subnet	Network	Mask	First host	Last host	Broadcast
SCCC LAN	11.8.0.0/23	255.255.254.0	11.8.0.1	11.8.1.254	11.8.1.255
DCS LAN	11.8.2.0/23	255.255.254.0	11.8.2.1	11.8.3.254	11.8.3.255
PHD LAN	11.8.4.0/24	255.255.255.0	11.8.4.1	11.8.4.254	11.8.4.255
TMA LAN	11.8.5.0/24	255.255.255.0	11.8.5.1	11.8.5.254	11.8.5.255
ESC LAN	11.8.6.0/24	255.255.255.0	11.8.6.1	11.8.6.254	11.8.6.255
ERU LAN	11.8.7.0/25	255.255.255.128	11.8.7.1	11.8.7.126	11.8.7.127
Central segment	11.8.7.128/29	255.255.255.248	11.8.7.129	11.8.7.134	11.8.7.135
SCCC – ERU	11.8.7.136/30	255.255.255.252	11.8.7.137	11.8.7.138	11.8.7.139
ERU – DCS	11.8.7.140/30	255.255.255.252	11.8.7.141	11.8.7.142	11.8.7.143
DCS – TMA	11.8.7.144/30	255.255.255.252	11.8.7.145	11.8.7.146	11.8.7.147
PHD – TMA	11.8.7.148/30	255.255.255.252	11.8.7.149	11.8.7.150	11.8.7.151
DCS – ESC	11.8.7.152/30	255.255.255.252	11.8.7.153	11.8.7.154	11.8.7.155

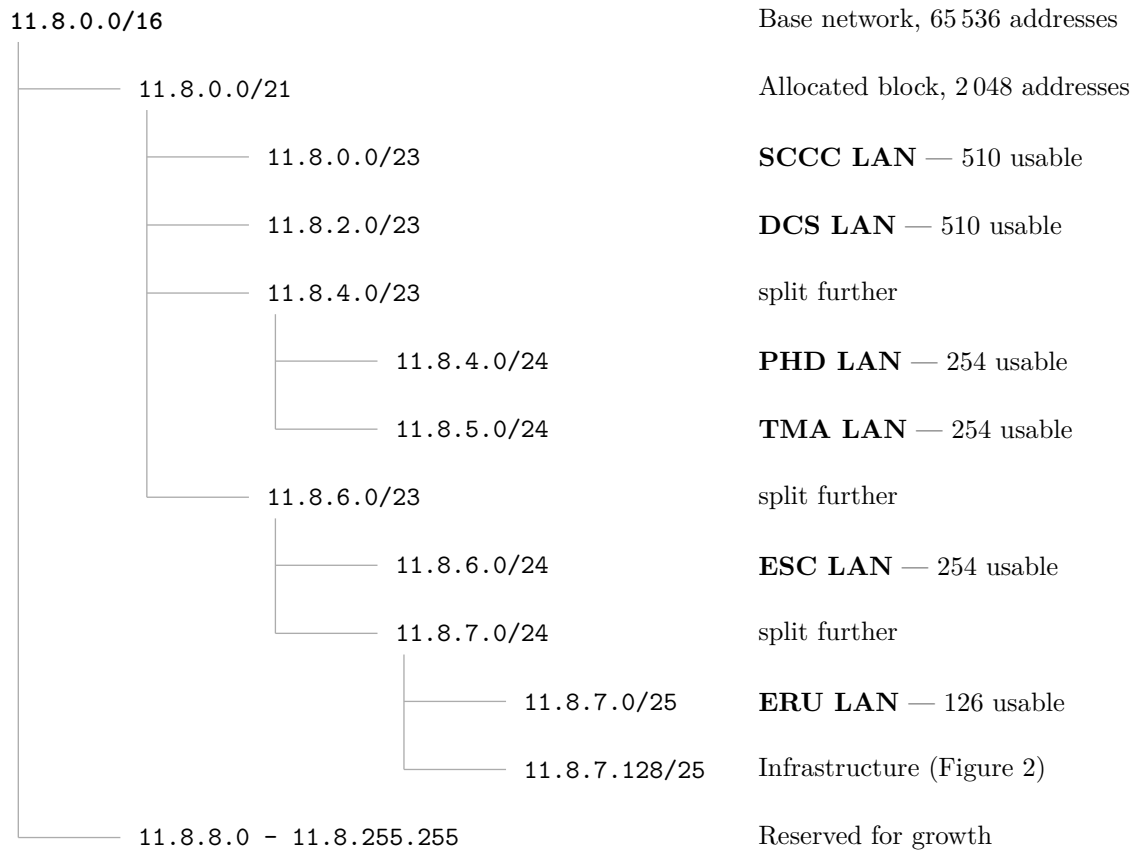


Figure 1: VLSM allocation tree for the six department LANs

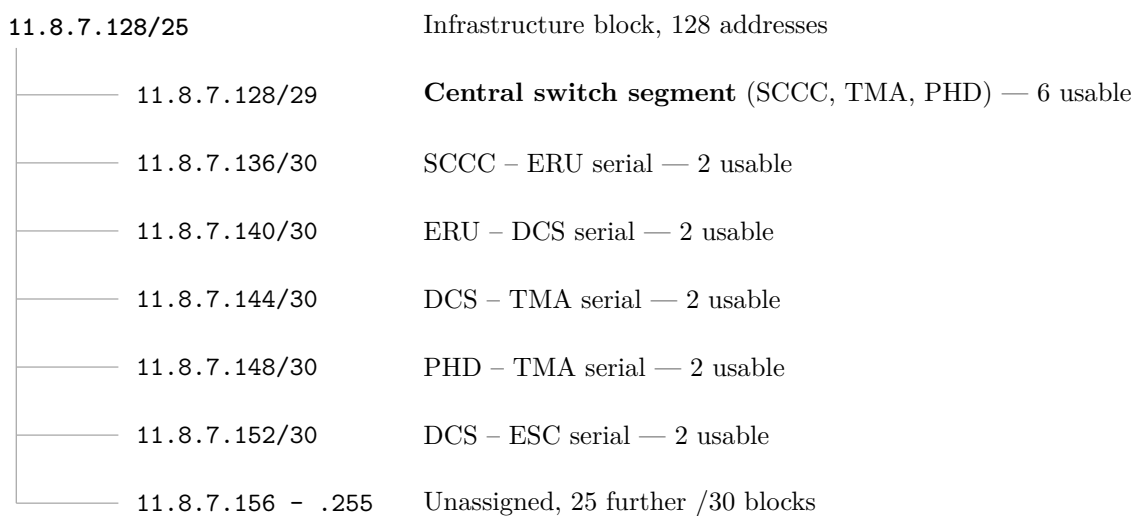


Figure 2: Expansion of the infrastructure block 11.8.7.128/25

3 IP Addressing Table

Routers, printers and servers use static addresses; all PCs receive their addresses dynamically. Every static host sits at the top of its LAN range and is excluded from the corresponding DHCP pool.

3.1 Router interfaces

Table 3: Router interface addressing

Router	Interface	Connects to	IP address	Mask
R-SCCC	Fa0/0	SCCC LAN switch	11.8.0.1	255.255.254.0
R-SCCC	Fa0/1	Central switch	11.8.7.129	255.255.255.248
R-SCCC	Se0/0/0	R-ERU Se0/0/0	11.8.7.137	255.255.255.252
R-TMA	Fa0/0	TMA LAN switch	11.8.5.1	255.255.255.0
R-TMA	Fa0/1	Central switch	11.8.7.130	255.255.255.248
R-TMA	Se0/0/0	R-DCS Se0/0/0	11.8.7.146	255.255.255.252
R-TMA	Se0/0/1	R-PHD Se0/0/0	11.8.7.149	255.255.255.252
R-PHD	Fa0/0	PHD LAN switch	11.8.4.1	255.255.255.0
R-PHD	Fa0/1	Central switch	11.8.7.131	255.255.255.248
R-PHD	Se0/0/0	R-TMA Se0/0/1	11.8.7.150	255.255.255.252
R-ERU	Fa0/0	ERU LAN switch	11.8.7.1	255.255.255.128
R-ERU	Se0/0/0	R-SCCC Se0/0/0	11.8.7.138	255.255.255.252
R-ERU	Se0/0/1	R-DCS Se0/1/0	11.8.7.141	255.255.255.252
R-DCS	Fa0/0	DCS LAN switch	11.8.2.1	255.255.254.0
R-DCS	Se0/0/0	R-TMA Se0/0/0	11.8.7.145	255.255.255.252
R-DCS	Se0/0/1	R-ESC Se0/0/0	11.8.7.153	255.255.255.252
R-DCS	Se0/1/0	R-ERU Se0/0/1	11.8.7.142	255.255.255.252
R-ESC	Fa0/0	ESC LAN switch	11.8.6.1	255.255.255.0
R-ESC	Se0/0/0	R-DCS Se0/0/1	11.8.7.154	255.255.255.252

Exactly one end of each serial link is the DCE and carries `clock rate 64000`. The five DCE ends are R-SCCC Se0/0/0, R-TMA Se0/0/1, R-ERU Se0/0/1, R-DCS Se0/0/0 and R-DCS Se0/0/1.

3.2 Servers and printers

Table 4: Static addressing for servers and printers

Dept.	Device	Role	IP address	Gateway
SCCC	SRV-DNS	Central DNS server	11.8.1.250	11.8.0.1
SCCC	SRV-WEB-SC	www.smartcity.gov.bd	11.8.1.251	11.8.0.1
SCCC	PRN-SCCC	Network printer	11.8.1.254	11.8.0.1
DCS	SRV-DHCP-DCS	DHCP for DCS and ESC	11.8.2.250	11.8.2.1

Dept.	Device	Role	IP address	Gateway
DCS	PRN-DCS	Network printer	11.8.3.254	11.8.2.1
PHD	PRN-PHD	Network printer	11.8.4.254	11.8.4.1
TMA	SRV-MAIL-TMA	mail.tma.gov.bd	11.8.5.250	11.8.5.1
TMA	PRN-TMA	Network printer	11.8.5.254	11.8.5.1
ESC	SRV-MAIL-ESC	mail.esc.gov.bd	11.8.6.250	11.8.6.1
ESC	PRN-ESC	Network printer	11.8.6.254	11.8.6.1
ERU	SRV-WEB-ER	www.emergency.gov.bd	11.8.7.120	11.8.7.1
ERU	SRV-DHCP-ERU	DHCP for ERU	11.8.7.121	11.8.7.1
ERU	PRN-ERU	Network printer	11.8.7.126	11.8.7.1

Every device in Table 4 uses 11.8.1.250 as its DNS server, including the two mail servers, which need it in order to relay cross-domain messages.

3.3 DHCP pools

Table 5: Dynamic address ranges

LAN	Served by	Pool range	Gateway	Delivery
SCCC	R-SCCC, pool SCCC-POOL	11.8.0.10 – 11.8.1.199	11.8.0.1	direct
TMA	R-SCCC, pool TMA-POOL	11.8.5.10 – 11.8.5.199	11.8.5.1	relay
PHD	R-SCCC, pool PHD-POOL	11.8.4.10 – 11.8.4.199	11.8.4.1	relay
DCS	SRV-DHCP-DCS	11.8.2.10 – 11.8.3.199	11.8.2.1	direct
ESC	SRV-DHCP-DCS	11.8.6.10 – 11.8.6.199	11.8.6.1	relay
ERU	SRV-DHCP-ERU	11.8.7.10 – 11.8.7.99	11.8.7.1	direct

All six pools hand out 11.8.1.250 as the DNS server. The relayed pools are reached through `ip helper-address`: TMA and PHD point at 11.8.7.129 (the SCCC router), and ESC points at 11.8.2.250 (the DCS server).

4 Configuration Commands

Every block below begins from user EXEC mode and assumes enable followed by configure terminal. Each router is finished with end and copy running-config startup-config.

4.1 R-SCCC

```

hostname R-SCCC
no ip domain-lookup
enable secret cisco123
banner motd #SBDC - Smart City Control Center - Authorized access only#
line console 0
  password cisco
  login
  logging synchronous
line vty 0 4
  password cisco
  login
exit
!--- interfaces
interface FastEthernet0/0
  description SCCC LAN
  ip address 11.8.0.1 255.255.254.0
  no shutdown
interface FastEthernet0/1
  description Central switch segment
  ip address 11.8.7.129 255.255.255.248
  no shutdown
interface Serial0/0/0
  description Serial to R-ERU DCE side
  ip address 11.8.7.137 255.255.255.252
  clock rate 64000
  no shutdown
exit
!--- DHCP server for SCCC, TMA and PHD
ip dhcp excluded-address 11.8.0.1 11.8.0.9
ip dhcp excluded-address 11.8.1.200 11.8.1.254
ip dhcp excluded-address 11.8.5.1 11.8.5.9
ip dhcp excluded-address 11.8.5.200 11.8.5.254
ip dhcp excluded-address 11.8.4.1 11.8.4.9
ip dhcp excluded-address 11.8.4.200 11.8.4.254
ip dhcp pool SCCC-POOL
  network 11.8.0.0 255.255.254.0
  default-router 11.8.0.1
  dns-server 11.8.1.250
  domain-name smartcity.gov.bd
ip dhcp pool TMA-POOL
  network 11.8.5.0 255.255.255.0
  default-router 11.8.5.1
  dns-server 11.8.1.250
ip dhcp pool PHD-POOL
  network 11.8.4.0 255.255.255.0
  default-router 11.8.4.1
  dns-server 11.8.1.250
exit
!--- dynamic routing
router rip
  version 2
  no auto-summary
  network 11.0.0.0
  passive-interface FastEthernet0/0
  passive-interface Serial0/0/0

```

```

timers basic 10 30 30 40
redistribute static metric 5
exit
!--- static routes toward the ERU / DCS / ESC side
ip route 11.8.7.0 255.255.255.128 11.8.7.138
ip route 11.8.7.140 255.255.255.252 11.8.7.138
ip route 11.8.2.0 255.255.254.0 11.8.7.138
ip route 11.8.7.144 255.255.255.252 11.8.7.138
ip route 11.8.7.152 255.255.255.252 11.8.7.138
ip route 11.8.6.0 255.255.255.0 11.8.7.138
!--- floating backup to the TMA LAN, distance 130
ip route 11.8.5.0 255.255.255.0 11.8.7.138 130

```

4.2 R-TMA

```

hostname R-TMA
no ip domain-lookup
enable secret cisco123
banner motd #SBDC - Traffic Management Authority#
line console 0
password cisco
login
logging synchronous
line vty 0 4
password cisco
login
exit
interface FastEthernet0/0
description TMA LAN
ip address 11.8.5.1 255.255.255.0
ip helper-address 11.8.7.129
no shutdown
interface FastEthernet0/1
description Central switch - PRIMARY PATH, failed during the demo
ip address 11.8.7.130 255.255.255.248
no shutdown
interface Serial0/0/0
description Serial to R-DCS - backup path
ip address 11.8.7.146 255.255.255.252
no shutdown
interface Serial0/0/1
description Serial to R-PHD DCE side - physical redundancy only
ip address 11.8.7.149 255.255.255.252
clock rate 64000
no shutdown
exit
router rip
version 2
no auto-summary
network 11.0.0.0
passive-interface FastEthernet0/0
passive-interface Serial0/0/0
passive-interface Serial0/0/1
timers basic 10 30 30 40
exit
!--- floating static routes via DCS, administrative distance 130
ip route 11.8.0.0 255.255.254.0 11.8.7.145 130
ip route 11.8.4.0 255.255.255.0 11.8.7.145 130
ip route 11.8.2.0 255.255.254.0 11.8.7.145 130
ip route 11.8.6.0 255.255.255.0 11.8.7.145 130
ip route 11.8.7.0 255.255.255.128 11.8.7.145 130

```

```
ip route 11.8.7.128 255.255.255.248 11.8.7.145 130
ip route 11.8.7.136 255.255.255.252 11.8.7.145 130
ip route 11.8.7.140 255.255.255.252 11.8.7.145 130
ip route 11.8.7.152 255.255.255.252 11.8.7.145 130
```

4.3 R-PHD

```
hostname R-PHD
no ip domain-lookup
enable secret cisco123
banner motd #SBDC - Public Health Department#
line console 0
  password cisco
  login
  logging synchronous
line vty 0 4
  password cisco
  login
exit
interface FastEthernet0/0
  description PHD LAN
  ip address 11.8.4.1 255.255.255.0
  ip helper-address 11.8.7.129
  no shutdown
interface FastEthernet0/1
  description Central switch segment
  ip address 11.8.7.131 255.255.255.248
  no shutdown
interface Serial0/0/0
  description Serial to R-TMA - physical redundancy only
  ip address 11.8.7.150 255.255.255.252
  no shutdown
exit
router rip
  version 2
  no auto-summary
  network 11.0.0.0
  passive-interface FastEthernet0/0
  passive-interface Serial0/0/0
  timers basic 10 30 30 40
```

4.4 R-ERU

```
hostname R-ERU
no ip domain-lookup
enable secret cisco123
banner motd #SBDC - Emergency Response Unit#
line console 0
  password cisco
  login
  logging synchronous
line vty 0 4
  password cisco
  login
exit
interface FastEthernet0/0
  description ERU LAN
  ip address 11.8.7.1 255.255.255.128
  no shutdown
```

```

interface Serial0/0/0
  description Serial to R-SCCC
  ip address 11.8.7.138 255.255.255.252
  no shutdown
interface Serial0/0/1
  description Serial to R-DCS DCE side
  ip address 11.8.7.141 255.255.255.252
  clock rate 64000
  no shutdown
exit
!--- reached through SCCC
ip route 11.8.0.0 255.255.254.0 11.8.7.137
ip route 11.8.4.0 255.255.255.0 11.8.7.137
ip route 11.8.7.128 255.255.255.248 11.8.7.137
ip route 11.8.7.148 255.255.255.252 11.8.7.137
!--- reached through DCS
ip route 11.8.2.0 255.255.254.0 11.8.7.142
ip route 11.8.6.0 255.255.255.0 11.8.7.142
ip route 11.8.5.0 255.255.255.0 11.8.7.142
ip route 11.8.7.144 255.255.255.252 11.8.7.142
ip route 11.8.7.152 255.255.255.252 11.8.7.142

```

4.5 R-DCS

```

hostname R-DCS
no ip domain-lookup
enable secret cisco123
banner motd #SBDC - Digital Citizen Service#
line console 0
  password cisco
  login
  logging synchronous
line vty 0 4
  password cisco
  login
exit
interface FastEthernet0/0
  description DCS LAN
  ip address 11.8.2.1 255.255.254.0
  no shutdown
interface Serial0/0/0
  description Serial to R-TMA DCE side
  ip address 11.8.7.145 255.255.255.252
  clock rate 64000
  no shutdown
interface Serial0/0/1
  description Serial to R-ESC DCE side
  ip address 11.8.7.153 255.255.255.252
  clock rate 64000
  no shutdown
interface Serial0/1/0
  description Serial to R-ERU
  ip address 11.8.7.142 255.255.255.252
  no shutdown
exit
!--- out of Serial0/0/0, toward TMA
ip route 11.8.5.0 255.255.255.0 Serial0/0/0
ip route 11.8.7.148 255.255.255.252 Serial0/0/0
!--- out of Serial0/0/1, toward ESC
ip route 11.8.6.0 255.255.255.0 Serial0/0/1
!--- out of Serial0/1/0, toward ERU and everything beyond it

```

```
ip route 11.8.0.0 255.255.254.0 Serial0/1/0
ip route 11.8.4.0 255.255.255.0 Serial0/1/0
ip route 11.8.7.0 255.255.255.128 Serial0/1/0
ip route 11.8.7.128 255.255.255.248 Serial0/1/0
ip route 11.8.7.136 255.255.255.252 Serial0/1/0
```

4.6 R-ESC

```
hostname R-ESC
no ip domain-lookup
enable secret cisco123
banner motd #SBDC - Education Service Center#
line console 0
  password cisco
  login
  logging synchronous
line vty 0 4
  password cisco
  login
exit
interface FastEthernet0/0
  description ESC LAN
  ip address 11.8.6.1 255.255.255.0
  ip helper-address 11.8.2.250
  no shutdown
interface Serial0/0/0
  description Serial to R-DCS
  ip address 11.8.7.154 255.255.255.252
  no shutdown
exit
!--- exactly one route, and nothing else
ip route 0.0.0.0 0.0.0.0 11.8.7.153
```

5 Server Configuration

The seven servers are configured through the Packet Tracer GUI rather than a CLI. The settings applied are recorded below.

5.1 DNS Server

Services → DNS → On, then add each record and press **Add**.

Table 6: DNS zone records

Name	Type	Address	Purpose
www.smartcity.gov.bd	A	11.8.1.251	SCCC website
www.emergency.gov.bd	A	11.8.7.120	ERU website
mail.tma.gov.bd	A	11.8.5.250	TMA mail client server field
mail.esc.gov.bd	A	11.8.6.250	ESC mail client server field
tma.gov.bd	A	11.8.5.250	cross-domain SMTP relay
esc.gov.bd	A	11.8.6.250	cross-domain SMTP relay
smartcity.gov.bd	A	11.8.1.251	convenience
emergency.gov.bd	A	11.8.7.120	convenience

5.2 Web Servers

On SRV-WEB-SC and SRV-WEB-ER: Services → HTTP → On, then edit `index.html`.

```
<!-- SRV-WEB-SC : www.smartcity.gov.bd -->
<html><body>
<h1>Smart City Control Center</h1>
<p>Smart Bangladesh Digital City - central services portal.</p>
<p>Server 11.8.1.251 - SCCC LAN 11.8.0.0/23</p>
</body></html>

<!-- SRV-WEB-ER : www.emergency.gov.bd -->
<html><body>
<h1>Emergency Response Unit</h1>
<p>24/7 emergency coordination - Smart Bangladesh Digital City.</p>
<p>Server 11.8.7.120 - ERU LAN 11.8.7.0/25</p>
</body></html>
```

5.3 DHCP Servers

Services → DHCP → On, fill the pool, then press **Add**.

Table 7: Server-based DHCP pools

Server	Pool	Gateway	Start IP	Mask	Max
SRV-DHCP-DCS	DCS-POOL	11.8.2.1	11.8.2.10	255.255.254.0	300
SRV-DHCP-DCS	ESC-POOL	11.8.6.1	11.8.6.10	255.255.255.0	190
SRV-DHCP-ERU	ERU-POOL	11.8.7.1	11.8.7.10	255.255.255.128	90

Every pool sets the DNS server field to 11.8.1.250. The ESC pool is matched by the relay agent address stamped by R-ESC, so its gateway field must read 11.8.6.1 and not 11.8.2.1.

5.4 Mail Servers

Services → EMAIL, switch SMTP and POP3 to On, set the domain name, then add each user.

Table 8: Mail server domains and user accounts

Server	Address	Domain	User	Full address
SRV-MAIL-TMA	11.8.5.250	tma.gov.bd	traffic1	traffic1@tma.gov.bd
SRV-MAIL-TMA	11.8.5.250	tma.gov.bd	admin	admin@tma.gov.bd
SRV-MAIL-ESC	11.8.6.250	esc.gov.bd	student	student@esc.gov.bd
SRV-MAIL-ESC	11.8.6.250	esc.gov.bd	teacher	teacher@esc.gov.bd

All four accounts use password `cisco123`.